

Testing of Hypothesis: Mean

Reading Objectives:

In this reading, you will learn about different models related to testing of hypothesis concerning mean. If the hypothesis involves mean, then we consider a suitable model accordingly.

Main Reading Section:

1. Testing of Hypothesis: One Mean—Large Sample

To test the claim that the population mean $H_0 : \mu = \mu_0$

We take the null hypothesis as $H_0 : \mu = \mu_0$.

The alternate hypothesis could be

$$H_1 : \mu \neq \mu_0 \text{ (two-sided) or } [H_1 : \mu < \mu_0 \text{ or } H_1 : \mu > \mu_0 \text{ ("one-sided")}]$$

Level of significance: α

Test statistic: $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ (σ known or large samples with $n \geq 30$)

Critical region: testing $\mu = \mu_0$ Critical region for (large sample)

Rejection Region	
Alternative Hypothesis	Reject Null hypothesis
$\mu < \mu_0$	$Z < -Z_\alpha$
$\mu > \mu_0$	$Z > Z_\alpha$
$\mu \neq \mu_0$	$Z > \frac{Z_\alpha}{2} \text{ or } Z < -\frac{Z_\alpha}{2}$

Note: If the population variance is unknown and the sample is large ($n \geq 30$), then we use z-test. In the test statistic, we replace the population variance with sample variance (s^2).

2. Testing of Hypothesis: One mean—Small Sample

The One Sample t-Test

Null hypothesis: $H_0: \mu = \mu_0$

Test statistic value: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Alternative Hypothesis

$$H_a: \mu > \mu_0$$

$$H_a: \mu < \mu_0$$

$$H_a: \mu \neq \mu_0$$

Rejection Region for a level α test

$$t \geq t_{\alpha, n-1} \text{ (upper-tailed)}$$

$$t \leq t_{-\alpha, n-1} \text{ (lower-tailed)}$$

$$\text{either } t \geq t_{\alpha/2, n-1} \text{ or } t \leq -t_{\alpha/2, n-1} \text{ (two-tailed)}$$

3. Testing of Hypothesis: Two Means—Large Sample

Null hypothesis: $H_0: \mu_1 - \mu_2 = d$

$$\text{Test statistic value: } z = \frac{\bar{x} - \bar{y} - d}{\sqrt{\frac{\sigma_1^2}{m} + \frac{\sigma_2^2}{n}}}$$

Alternative Hypothesis

$$H_a: \mu_1 - \mu_2 > d$$

$$H_a: \mu_1 - \mu_2 < d$$

$$H_a: \mu_1 - \mu_2 \neq d$$

Rejection Region for Level α Test

$$z \geq z_\alpha \text{ (upper-tailed)}$$

$$z \leq -z_\alpha \text{ (lower-tailed)}$$

$$\text{either } z \geq z_{\alpha/2} \text{ or } z \leq -z_{\alpha/2} \text{ (two-tailed)}$$

4. Testing of Hypothesis: Two Means—Small Sample

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}},$$

where

$$s_p^2 = \frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}$$